Full-Stack Development

PRACTICAL NO. 1(A)

Aim:-Install Node.js and visual Studio code ,and verify installation.

Step1: Install node.js
1.Visit the official Node.js website:
Node.js Download Page.
2.Download the LTS(Long Term Support) version.
3.Run the installer and Follow the installation wizard.
4.Keep all default setting and click Next until installation completes.\

Step 2: Install visual studio code:
1.Visit: Visual Studio Code
2.Download and Install VS Code.
3.Open VS Code after installation.

Step 3: Verify installation:
Open Command Prompt Or Terminal and type:
Node -v

Output:
Npm -v

Output:
10.5.0

PRACTICAL NO. 1(B)

Aim:- Create a simple Node.js File and run it.

Step 1: Create a Folder
Create a folder named:
NodeJS_practical

Step 2: Open Folder in VS Code
1.Open Vs Code
2.Select FIle >> Open Folder
3.Choose NOdeJS_practical

Step 3: Create a File
Create a file named:
app.js

Step 4: Write Code
console.log("Hello world");

Step 5: Run the Program
Open Terminal in VS Code:
Node app.js

Output:
Hello world

PRACTICAL NO. 1(C)

Aim:- Write a program demonstrating basic javascript command.

Code:-

```
console.log("Welcome to node js");
let name="John";
let age=20;
console.log("Name:",name);
console.log("Age:",age);
let num1=15;
let num2=5;
console.log("First Number:",num1);
console.log("Second Number:",num2);
console.log("Addition:",num1+num2);
console.log("Substraction:",num1-num2);
console.log("Multiplication:",num1*num2);
console.log("Division:",num1/num2);
console.log("Modulus:",num1%num2);
```

Output:-

```
Welcome to node js
Name: John
Age: 20
First Number: 15
Second Number: 5
Addition: 20
Substraction: 10
Multiplication: 75
Division: 3
Modulus: 0
```

PRACTICAL NO. 1(D)

Aim:- Create and run a simple program using variable and functions.

Code:-

```
let name= "Adwait";
let marks=85;
function displayStudent(studentName, studentMarks)
{
console.log("Student Name:",studentName);
console.log("Marks:",studentMarks);
}
displayStudent(name, marks);
```

Output:-

Student Name: Adwait
Marks: 85

(Sum of Two Numbers)

Code:-

```
let num1=10;
let num2=20;

function add(a,b){
   return a+b;
}

let result=add(num1,num2);

console.log("First Number:",num1);
console.log("Second Number:",num2);
console.log("Sum:",result);
```

Output:-

First Number: 10
Second Number: 20
Sum: 30

PRACTICAL NO. 1(E)

Aim:- Demonstrate the use of operator and control statement in javascript.

(Operators)

Code:-

```
let a=15;
let b=5;

//Arithmetic Operator
console.log("Addition:",a+b);
console.log("Substraction:",a-b);
console.log("Multiplication:",a*b);
console.log("Divison:",a/b);
console.log("Modulus:",a%b);

//Comparision Operator
console.log("Is a greater than b?",a>b);

//Logical Operator
console.log("a>10 && b<10",a>10 && b<10);
```

Output:-

Addition: 20
Substraction: 10
Multiplication: 75
Divison: 3
Modulus: 0
Is a greater than b? true
a>10 && b<10 true

(If - Else Statement)

Code:-

```
let marks=80;
if(marks>=50){
   console.log("Result:Pass");
}else{
   console.log("Result:Fail");
}
```

Output:-

Result: Pass

(Switch Case)

Code:-

```
let day=3;

switch(day){
   case 1:
      console.log("Monday");
      break;
   case 2:
      console.log("Tuesday");
      break;
   case 3:
      console.log("Wednesday");
      break;
   case 4:
      console.log("Thursday");
      break;
   default:
      console.log("Invalid Day!");
}
```

Output:-

Wednesday

PRACTICAL NO. 1(F)

Aim:- Write a program using loops and template literals.

(For Loop)

Code:-

```
let student="Amit";
console.log(`Welcome,${student}!`);

for(let i=1;i<=5;i++)
{
   console.log(`Iteration ${i}:${student} is learning Node.js`);
}
```

Output:-

Iteration 1:Amit is learning Node.js

Iteration 2:Amit is learning Node.js

Iteration 3:Amit is learning Node.js

Iteration 4:Amit is learning Node.js

Iteration 5:Amit is learning Node.js

(While Loop)

Code:-

```
let number=1;

while(number<=5)
{
   console.log(`The square of ${number} is ${number*number}`);
   number++;
}
```

Output:-

The square of 1 is 1
The square of 2 is 4
The square of 3 is 9
The square of 4 is 16
The square of 5 is 25

(Do-While Loop)

Code:-

```
let count=1;

do{
   console.log(`Count value is ${count}`);
   count++;
} while(count<=5);
```

Output:-

```
Count value is 1
Count value is 2
Count value is 3
Count value is 4
Count value is 5
```

(Multiplication Table)

Code:-

```
let num=7;

console.log(`Multiplication Table of ${num}`);

for(let i=1;i<=10;i++)
{
   console.log(`${num}x${i}=${num*i}`);
}
```

Output:-

```
7x1=7
7x2=14
7x3=21
7x4=28
7x5=35
7x6=42
7x7=49
7x8=56
7x9=63
7x10=70
```

(Sum of 10 Numbers)

Code:-

```
let sum=0;

for(let i=1;i<=10;i++)
{
   sum+=i;
}
console.log(`The sum of numbers from 1 to 10 is ${sum}.`);
```

Output:-

The sum of numbers from 1 to 10 is 55.

(Eligibility Criteria for Voting)

Code:-

```
let age=20;

if(age>=18){
   console.log("Person is eligible to vote.");
}
else{
   console.log("Person is not eligible to vote.")
}
```

Output:-

Person is eligible to vote.

(Grade Checker Using Switch Case)

Code:-

```
let marks = 85;

switch (true) {
   case (marks>=90 && marks<=100):
      console.log("Grade: A+");
      break;

   case (marks>=80):
      console.log("Grade: A");
      break;

   case (marks>=70):
      console.log("Grade: B");
      break;

   case (marks>=60):
      console.log("Grade: C");
      break;

   case (marks>=50):
      console.log("Grade: D");
      break;

   case (marks>=0):
      console.log("Grade: F");
      break;

   default:
      console.log("Invalid marks");
}
```

Output:-

Grade: A